ETH zürich

Mobile Computing

Handling Data in Mobile Computing

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Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

Handling Data in Mobile Computing

- What are the mechanisms by which information is processed and represented in digital systems?
- What techniques are employed for data encoding, compression, and optimization in communication systems?
- How are concepts of data entropy and redundancy applied in communication systems?

Data, Information, Knowledge

- Data refers to raw, unprocessed facts, figures, or symbols that do not carry meaning on their own
  - Numbers, text, measurements - lacks context or interpretation
- Information is processed data that has been organized, or interpreted so that it has meaning
  - Results from analyzing and understanding data
  - Useful for decision-making, or conveying knowledge
- Knowledge allows a system to act on information
  - Applying rules, patterns, or insights gained from information
  - To make informed decisions or predictions

Example: Tracking Heart Rates

- Data: Raw heart rate readings
  - Values like "80 bpm," or "90 bpm", recorded at specific intervals
- Information: Processing and analyzing
  - For example: "The average resting heart rate over the past week is 10% higher than the normal range"
  - This provides meaningful insights
- Knowledge: Advising to take specific actions
  - For example, recognizing that a consistently elevated resting heart rate might indicate a potential health issue
  - Suggesting that a user may need to react or rest more

Analogy

- Data = words in a dictionary (all possible words)
- Information = sentences in a newspaper (words with meaning)
- Knowledge = a journalist’s understanding (able to explain, connect, and create new sentences)

Data: the Raw Material

- Facts and symbols without context or interpretation
- Training corpus in an LLM: billions of words, sentences, code snippets, etc., scraped from books, websites, and other sources

Example:

- "01000001" (bits), "apple", "3.14", a sentence from Wikipedia
- By itself, it is just raw text, no meaning yet to the model

Information: Data with Structure and Meaning

- Processed data that answers a specific question or has a purpose
- During LLM tokenization, statistical relationships between words (like “wireless” is often near “communication”) become information because they carry meaning

Example:

- The token "Paris is the capital of France" is information as it is organized and has meaning

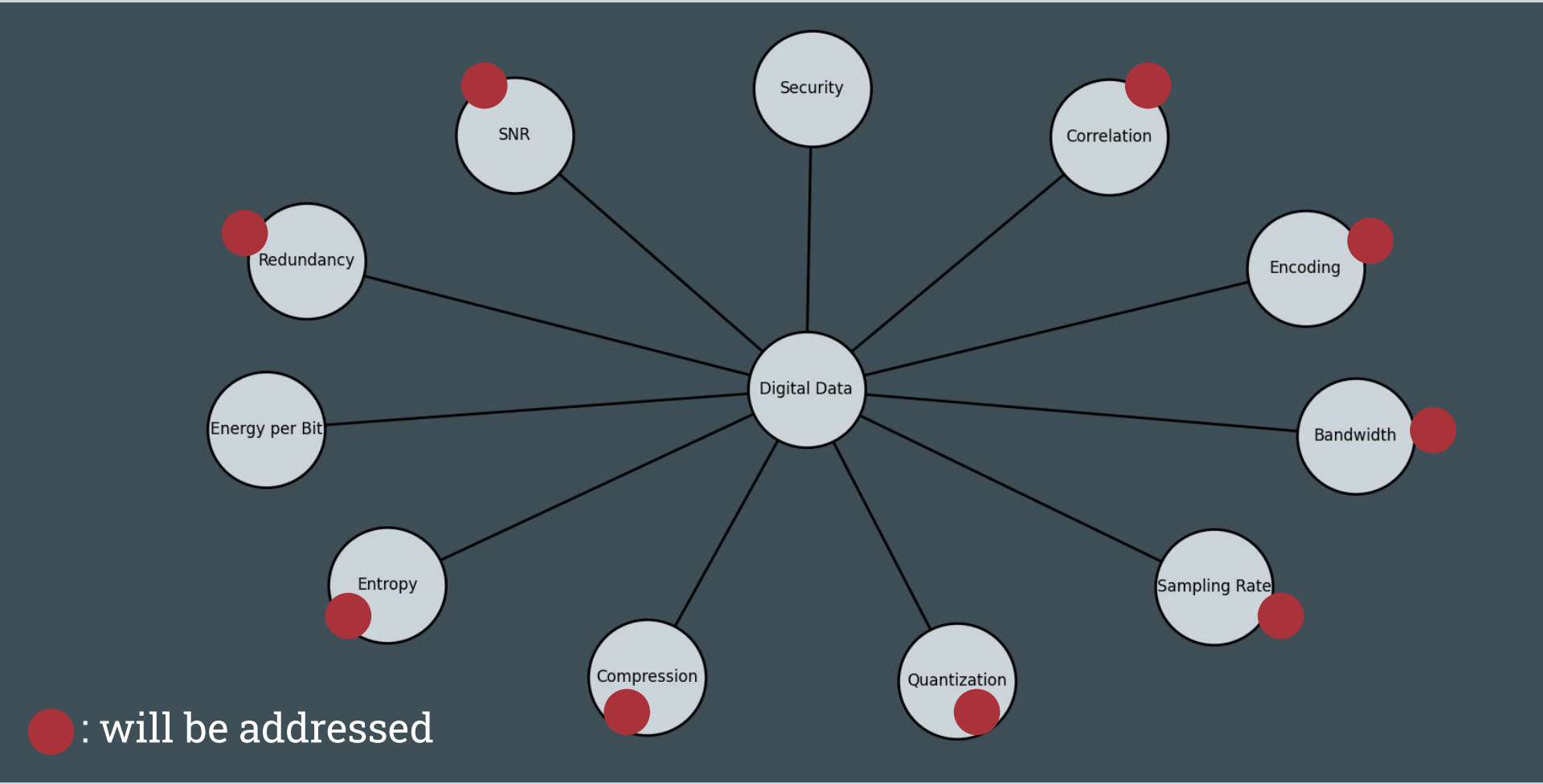
Knowledge: Generalized Understanding

- Information that can be applied to reason, predict, or explain
  - After LLM training, a model has internalized probabilistic models of facts.
  - This is its knowledge, as it provides the ability to infer new connections from patterns

Example:

- Even if the sentence “The federal center of Switzerland is Bern” did not appear in training data, a foundational LLM should be able to generate it

Digital Data - Key Concepts

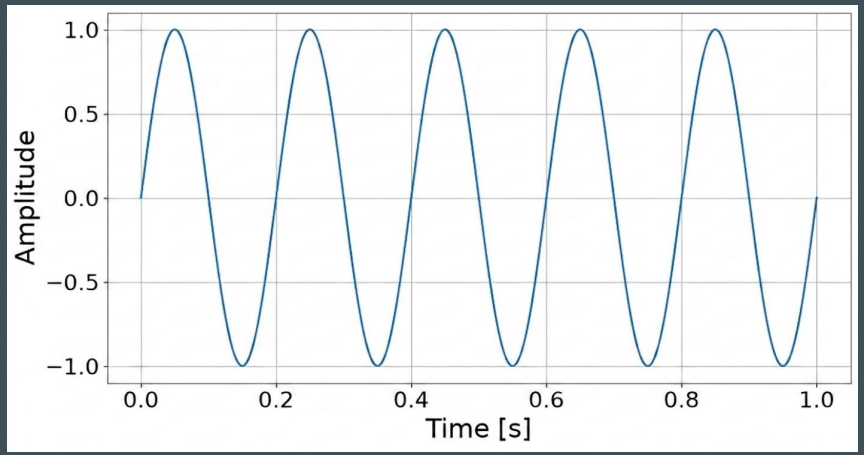


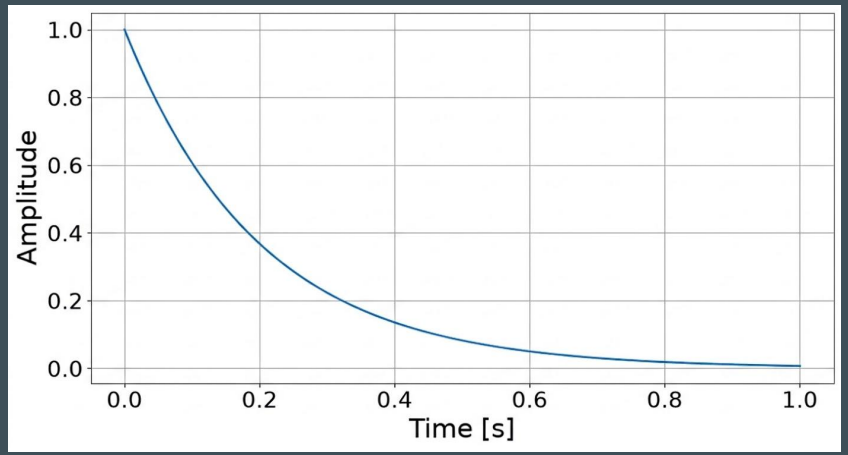
Analog Signals and Digital Data

- Information needs a representation before it can be stored, processed, or transmitted
- Analog values are continuous; digital data uses discrete symbols or steps
- The chosen representation affects noise sensitivity, storage, copying, and processing

Analog Signals

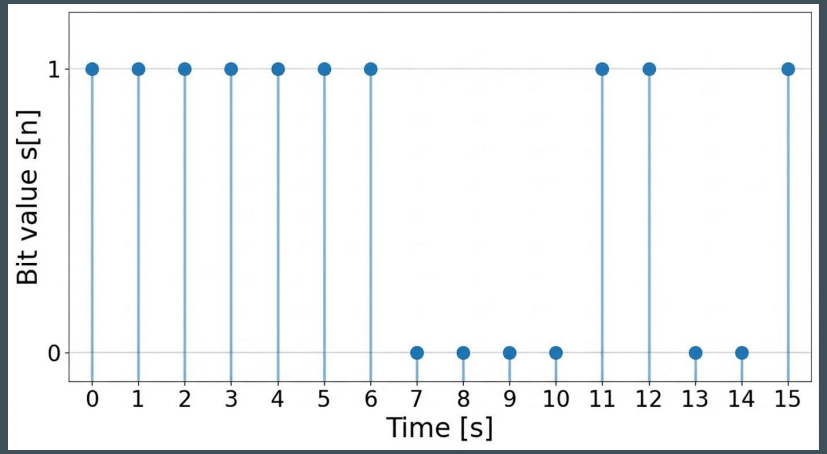
- Continuous-valued quantities can represent voltage, radio waves, light, sound, or temperature
- Physical transmission media naturally carry continuous signals

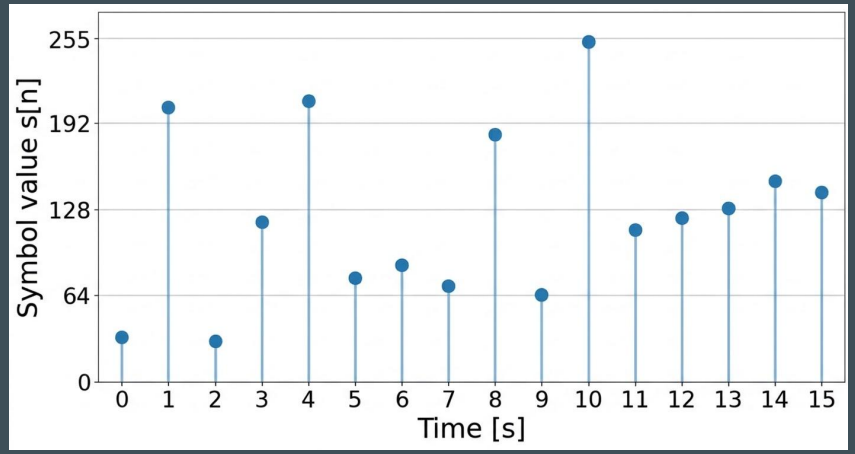




Digital Data

- Discrete symbols represent numbers, text, images, sensor readings, and audio
- Digital data can be stored, copied, compressed, and protected against errors reliably





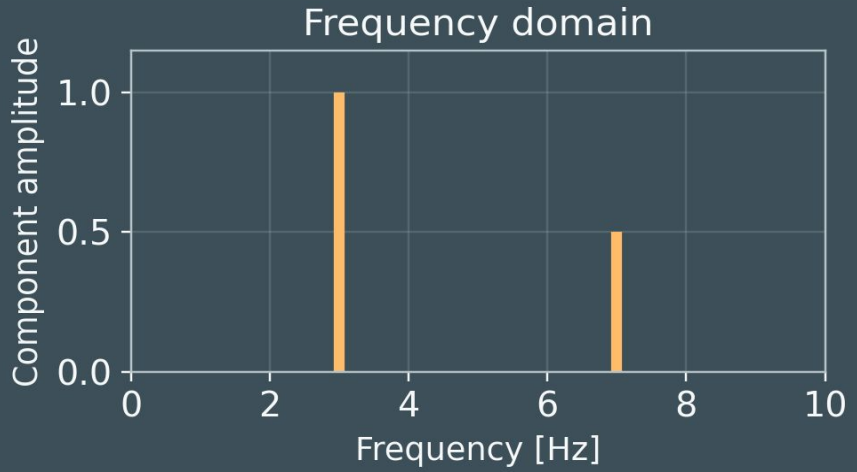
Exercise: Analog and Digital

- Work through 01.1_Analog_Signals_and_Digital_Data.ipynb and complete the exercise
- Save the completed notebook and upload it to Moodle

One signal, two views

- The waveform shows when values change. The spectrum shows its frequency components.





A frequency component needs a frequency, an amplitude, and a phase.

Build a waveform from sinusoids

- Change one component at a time to see what amplitude, frequency, and phase control.

$$s_m(t) = A_m \sin(2\pi f_m t + \phi_m)$$

$$x(t) = \sum_m s_m(t)$$

Amplitude sets strength. Frequency sets cycles per second. Phase sets position within a cycle.

Measure the frequency components

- Multiply the waveform by reference oscillations and integrate over time.

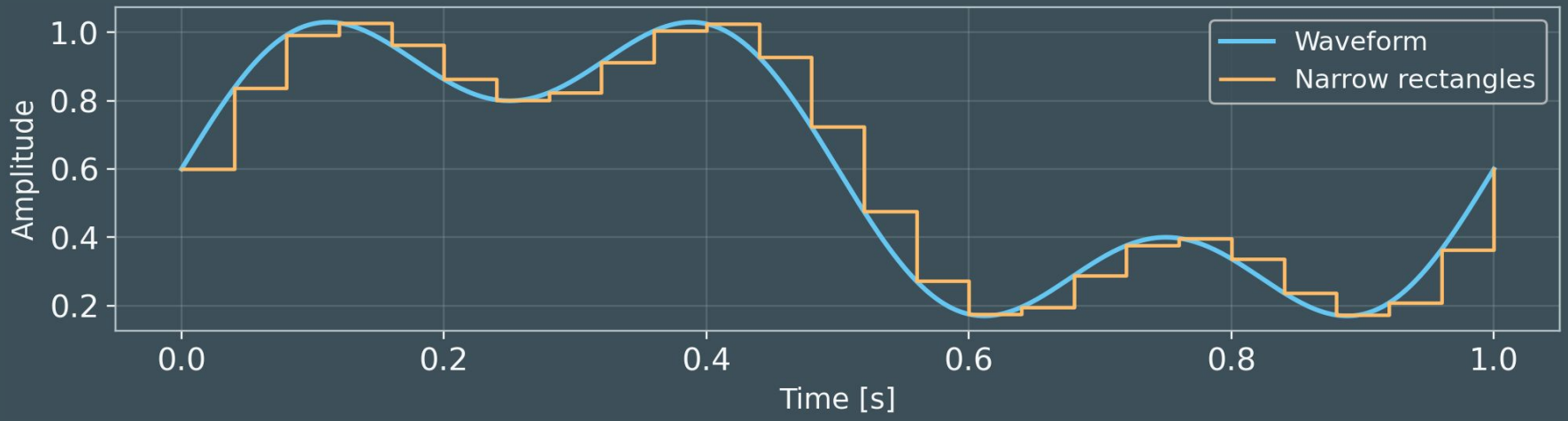
$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft}\, dt$$

$$e^{-j2\pi ft} = \cos(2\pi ft) - j\sin(2\pi ft)$$

$j^2 = -1$. The complex coefficient X(f) contains magnitude and phase. Time is in seconds; frequency is in hertz.

Why can sinusoids represent a signal?

- Approximate a waveform by narrow rectangles; each rectangle can itself be built from sinusoids.



Add all contributions and combine those at the same frequency. More bandwidth captures finer detail.

Reconstruction needs phase

- The inverse transform combines the complex frequency components.

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft}\, df$$

$$\sin(\omega t) + \sin(\omega t + \pi) = 0$$

Magnitudes alone are insufficient: phases determine where components reinforce or cancel.

Same magnitudes, different waveform

- Changing a component phase changes the sum without moving the spectral amplitude peaks.





A magnitude spectrum does not tell us where a pulse or waveform feature occurs in time.

Seven harmonics: recognizable, not exact

- For a periodic target, harmonic frequencies are integer multiples of the fundamental.

$$f_m = m f_0, \quad f_0 = 1 \text{ Hz}$$

$$\hat{x}(t) = c_0 + \sum_{m=1}^{7} A_m \sin(2\pi m f_0 t + \phi_m)$$

c_0 is the mean. The highest retained frequency is 7 Hz. Sharp edges require frequencies beyond any finite cutoff.

Exercise: Time and frequency

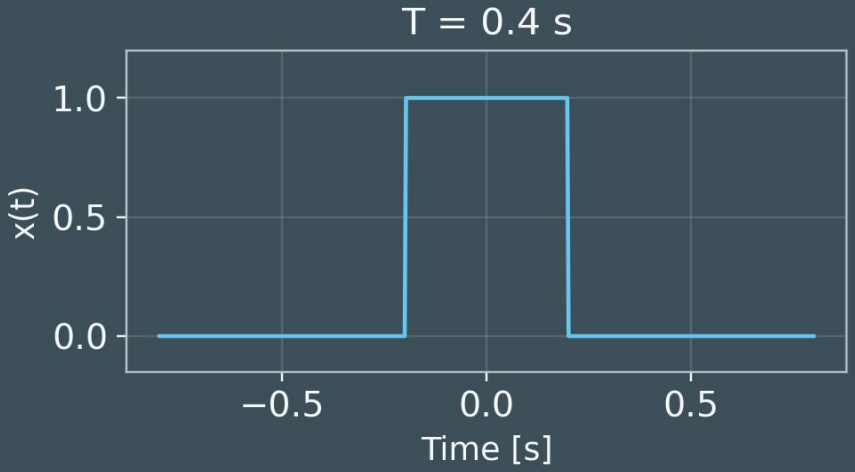
- Work through the notebook and complete its exercise.

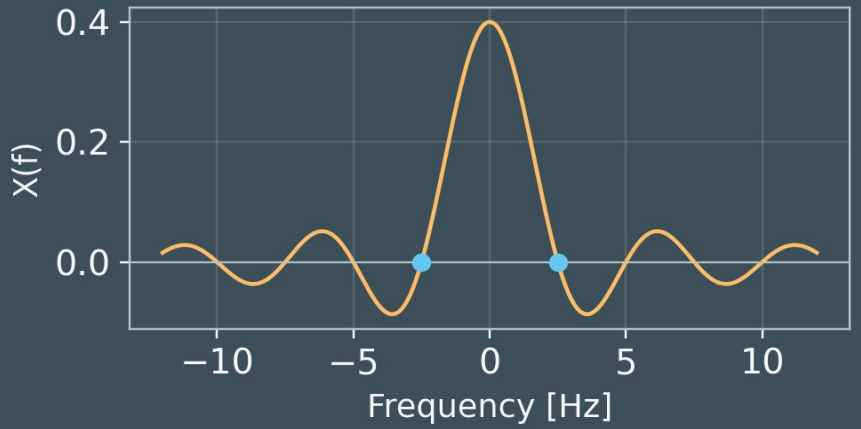
  01.2_Representing_Signals_in_Time_and_Frequency_Domain.ipynb

- Adjust the deliberately incorrect presets. Compare the corrected sums with their targets and explain the remaining differences.
- Save the completed notebook and upload it to Moodle.

A short pulse has a broad spectrum

- A rectangle gives us a concrete bridge from a time-domain shape to its frequency content.





The pulse is finite in time; its sinc spectrum extends to arbitrarily high frequencies.

Derive the rectangular-pulse transform

- For a unit-height pulse centered at zero, only the interval −T/2 to T/2 contributes.

$$x(t) = \mathrm{rect}(t/T)$$

$$X(f) = \int_{-T/2}^{T/2} e^{-j2\pi ft}\, dt = \frac{e^{-j\pi fT} - e^{j\pi fT}}{-j2\pi f}$$

Inside the pulse x(t)=1; outside it x(t)=0. The Fourier reconstruction takes the value 1/2 at each edge.

Euler’s formula explains the sine

- Subtract the two complex exponentials: their cosine terms cancel.

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$e^{-j\theta} - e^{j\theta} = -2j\sin\theta$$

$$X(f) = \frac{\sin(\pi f T)}{\pi f} = T\,\mathrm{sinc}(fT)$$

We use normalized sinc(u)=sin(πu)/(πu). At f=0, use the limit: X(0)=T, the pulse area.

State which bandwidth you mean

- An ideal rectangle has infinite absolute bandwidth, even though its main lobe has finite width.

$$f_{\text{first zero}} = \pm\frac{1}{T}$$

$$B_{\text{first null}} = \frac{1}{T}, \qquad W_{\text{main lobe}} = \frac{2}{T}$$

T=0.4 s gives zeros at ±2.5 Hz and a 5 Hz main-lobe width. Halving T doubles both finite measures.

DFT is the transform; FFT is the algorithm

- A computer represents a finite sample block using discrete frequency coefficients.

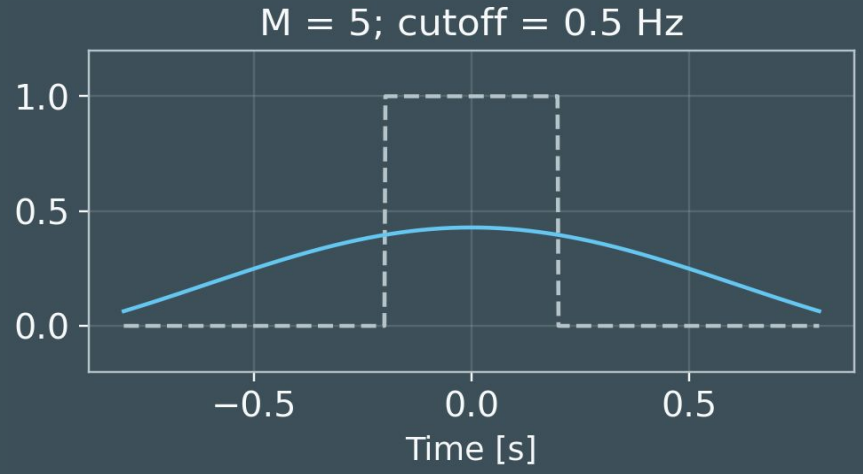
$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

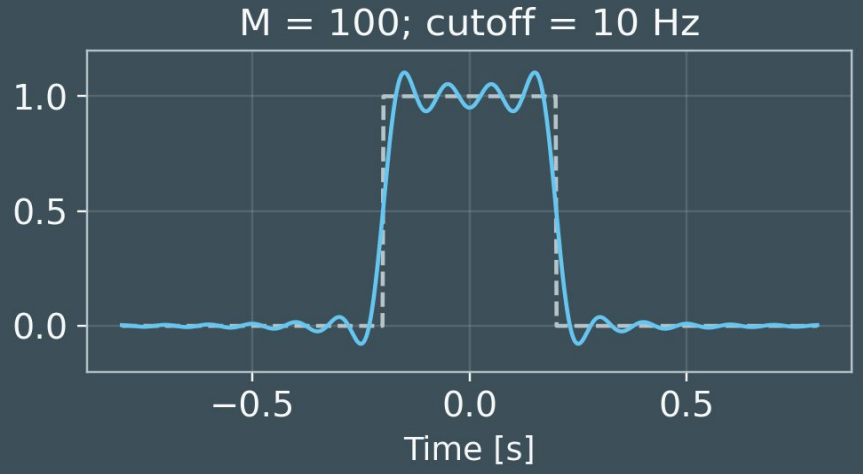
$$\Delta f = \frac{F_s}{N} = \frac{1}{T_{\text{obs}}}$$

The DFT treats the block as periodic. Keeping every coefficient recovers every sample; the FFT computes the same DFT efficiently.

Finite bandwidth rounds the edges

- Keep DC and M positive/negative harmonic pairs, then reconstruct.





Δf=0.1 Hz: M=5 gives 0.5 Hz; M=100 gives 10 Hz. More components sharpen the transition, but Gibbs overshoot remains near the jump.

A zero-order hold is a row of pulses

- Hold each sample value for one sampling interval to obtain a staircase waveform.

$$x_{\mathrm{ZOH}}(t) = \sum_n x[n]\,\mathrm{rect}\left(\frac{t-(n+1/2)T_s}{T_s}\right)$$

Each pulse starts at nT_s, lasts T_s, and has height x[n]. This is a reconstruction rule, not an inherent part of a sample sequence.

Delay changes phase, not magnitude

- Shift the whole waveform by the same time interval.

$$x_d(t) = x(t - t_0)$$

$$X_d(f) = X(f)e^{-j2\pi f t_0}, \qquad |X_d(f)| = |X(f)|$$

Magnitude-based bandwidth stays the same. Moving individual pulses relative to one another can change the magnitude spectrum.

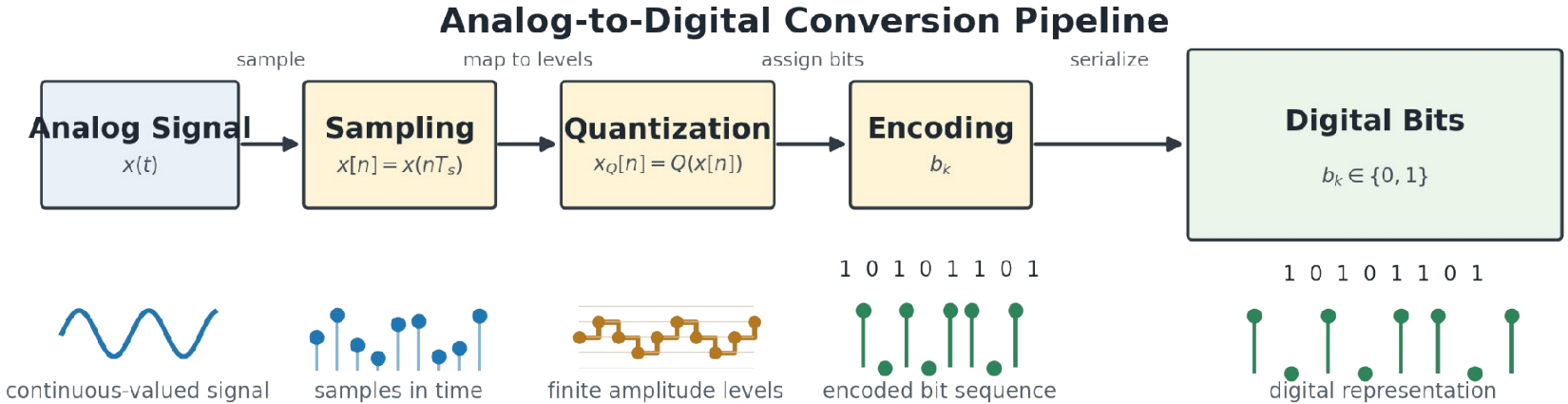
Exercise: Rectangular pulses

- Work through the notebook and complete its exercise.

  01.3_Fourier_Approximation_of_a_Rectangular_Pulse.ipynb

- Relate pulse duration to bandwidth. Explore finite-bandwidth reconstruction, zero-order hold, and the effect of a common time shift.
- Save the completed notebook and upload it to Moodle.

Sampling, Quantization, and Encoding



From waveform to digital samples

- An ADC separates three operations that solve different representation problems.

Operation	Input → output	What becomes discrete?
Sampling	x(t) → x[n]	Time positions
Quantization	x[n] → xQ[n]	Amplitude levels
Encoding	Level index → binary word	Symbol representation

More samples improve time resolution. More bits provide finer amplitude resolution.

Sampling selects time positions

- A sample records the waveform at one instant; it does not specify values between samples.

$$x[n] = x(nT_s), \qquad f_s = \frac{1}{T_s}$$

$$t_n = nT_s$$

T_s is in seconds; f_s is in samples per second. A zero-order hold is one possible way to represent the sequence as a waveform.

How fast must we sample?

- For a band-limited signal, the highest wanted frequency must lie below the Nyquist frequency.

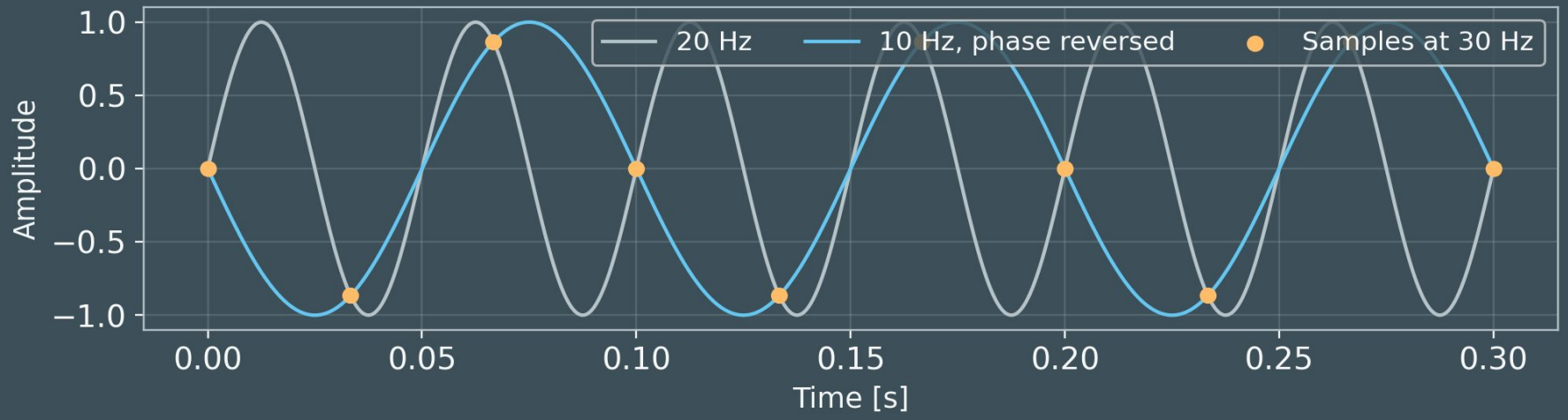
$$f_s > 2f_{\max}, \qquad f_{\text{Nyquist}} = \frac{f_s}{2}$$

$$x(t) = 0.5\sin(2\pi 6t) + 0.5\sin(2\pi 20t)$$

Here fmax=20 Hz, so choose f_s>40 Hz. Exact reconstruction assumes exact uniform samples over all time and ideal reconstruction.

Aliasing: different signals, identical samples

- At 30 samples/s, a 20 Hz sine has the same samples as a phase-reversed 10 Hz sine.



The samples alone cannot tell which waveform was present. More bits cannot resolve this ambiguity.

Filtering needs room before sampling

- An anti-aliasing filter suppresses unwanted frequencies before the ADC.

$$f_{\max} < \frac{f_s}{2}$$

$$f_k = k\frac{f_s}{N}$$

Real filters need a transition band. A larger sampling margin helps filtering but increases data volume. A DFT spectrum uses bins spaced f_s/N apart.

Quantization chooses an amplitude level

- The notebook’s uniform quantizer includes both endpoints of [0,1].

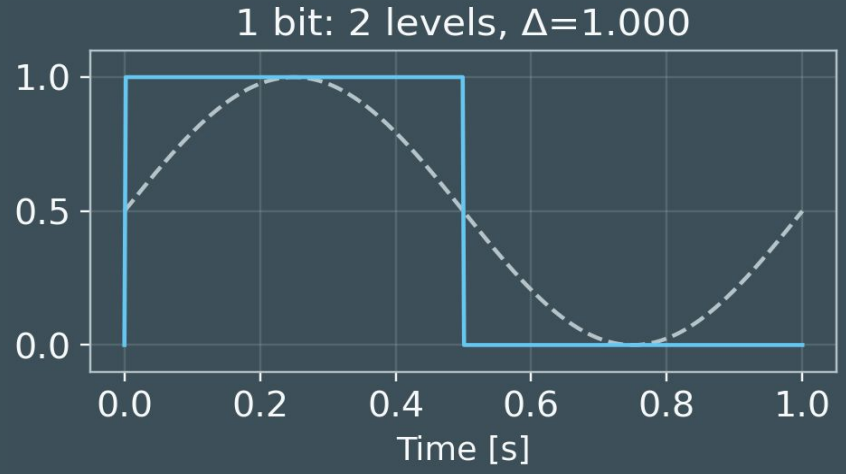
$$L = 2^B, \qquad \Delta = \frac{1}{2^B - 1}$$

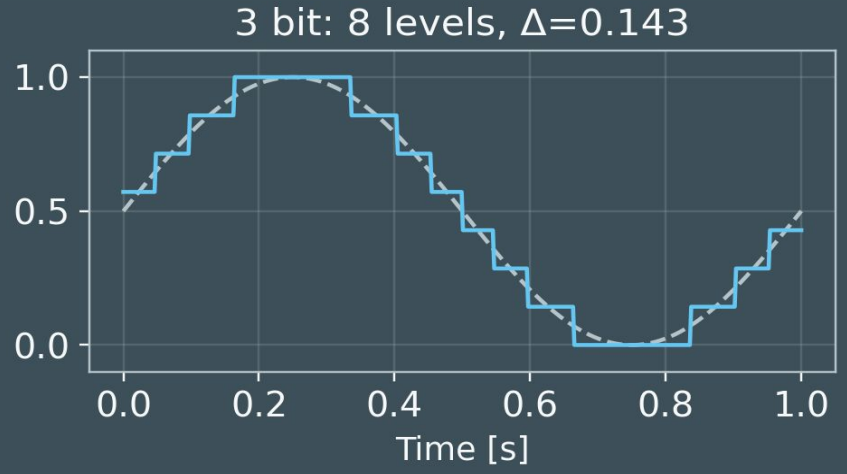
$$Q(x[n]) = \Delta \,\mathrm{round}\left(\frac{x[n]}{\Delta}\right)$$

B is bits per sample; L is the number of levels here. Later, L denotes average codeword length: read the definition in context.

More bits reduce rounding error

- For inputs within the representable range, nearest-level rounding stays within half a step.





$e[n]=Q(x[n])-x[n]$, with $|e[n]|\le\Delta/2$. This bound does not cover clipping outside the allowed range.

Three bits identify eight levels

- Quantize an input value, then encode the level index rather than the amplitude itself.

$$x[n] = 0.6, \quad \Delta = 1/7, \quad i = \mathrm{round}(0.6 \cdot 7) = 4$$

$$Q(x[n]) = 4/7 \approx 0.5714, \quad e[n] \approx -0.0286$$

Index 4 is transmitted as 100. With B bits per sample and one channel, the raw data rate is $R = f_s B$ bit/s, before overhead.

Exercise: Sampling, quantization, encoding

- Work through the notebook and complete its exercise.

  01.4_Converting_Analog_Signals_to_Digital_Time_Series.ipynb

- Choose a sampling rate and bit depth, interpret aliasing, and calculate quantization error and raw data rate.
- Save the completed notebook and upload it to Moodle.

Three stored bits: how much information?

- Treat each complete word 000 through 111 as one source symbol.

Quantity	Meaning	Unit
I(x)	Surprise of one outcome	bit
H(X)	Average surprise of a source symbol	bit/source symbol
L	Average encoded length	bit/source symbol

A symbol's label, its probability, and the number of bits used to encode it are different quantities.

Rare outcomes carry more information

- Self-information assigns a larger value to a less probable event.

$$I(x) = -\log_2 p(x)$$

$$p = 1 \Rightarrow I = 0, \quad p = 1/8 \Rightarrow I = 3, \quad p = 1/40 \Rightarrow I \approx 5.322$$

The unit is bit because the logarithm has base 2. Information content can be fractional; it is not an individual codeword's length.

Entropy is the probability-weighted average

- Weight each outcome's information content by how often it occurs.

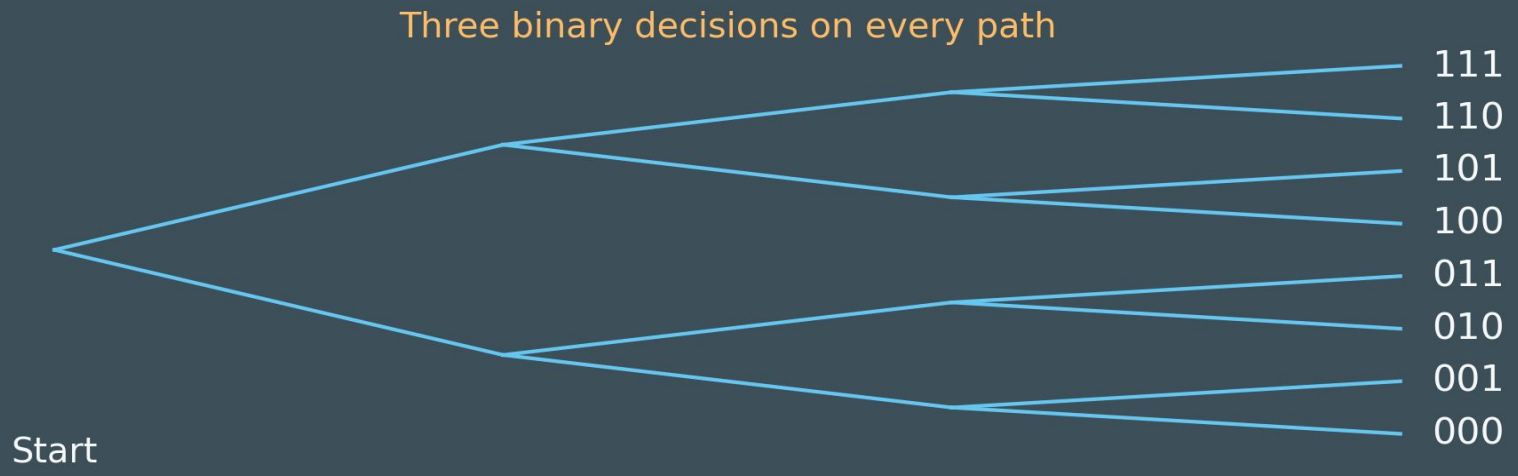
$$H(X) = \sum_i p_i I(x_i) = -\sum_i p_i \log_2 p_i$$

$$\sum_i p_i = 1, \qquad 0 \log_2 0 := 0$$

Symbol entropy considers one symbol alone. It does not capture predictable ordering between symbols.

Eight equally likely symbols

- Three balanced yes/no decisions identify a symbol.



Each p=1/8 gives I=3 bit. The weighted average H=3 bit/symbol equals the fixed-length code's L=3.

Frequent symbols can be identified sooner

- Use the same eight labels with unequal probabilities.

Source symbol	Probability	Information [bit]
000	1/2	1
001	1/4	2
010	1/8	3
Each of 011…111	1/40	5.322

Half of all symbols are 000: asking about it first often finishes the identification after one decision.

A decodable variable-length code

- No complete codeword is a prefix of another: the receiver can find symbol boundaries.

Symbol	p	Codeword	Symbol	p	Codeword
000	1/2	0	100	1/40	11101
001	1/4	10	101	1/40	11110
010	1/8	110	110	1/40	111110
011	1/40	11100	111	1/40	111111

The original three-bit source symbol and its transmitted codeword are different representations of the same value.

Compare entropy with actual code length

- Frequent symbols use short words; rare ones pay for the longer words.

$$L = \tfrac{1}{2}(1) + \tfrac{1}{4}(2) + \tfrac{1}{8}(3) + \tfrac{1}{40}(3 \cdot 5 + 2 \cdot 6) = 2.05$$

$$H(X) \approx 2.04024 \text{ bit/symbol} < L$$

For 1,000 symbols in these proportions: 3,000 → 2,050 bits, a 31.7% saving, excluding headers and the code description.

Identical counts do not imply identical structure

- Sixteen three-bit symbols can have maximum symbol entropy and still follow a repeating rule.

Pattern	$\hat{H}$ [bit/symbol]
000 repeated 16 times	0
000 111 alternating	1
000 001 … 111, repeated twice	3
Same eight symbols, each twice, shuffled	3

$\hat{H}$ uses observed frequencies. Order requires a model with context. Sixteen observations do not establish true randomness or an exact compressed size.

Exercise: Information content and entropy

- Work through the notebook and complete its exercise.

  01.5_Entropy.ipynb

- Calculate surprise, entropy, and code length. Compare ordered and shuffled symbols, then check your calculations with the notebook function.
- Save the completed notebook and upload it to Moodle.

Two opportunities for compression

- A representation can waste bits because it ignores probabilities or predictable symbol order.

Structure	What the model uses	Example
Unequal frequencies	Probability of each symbol	000 occurs half the time
Sequential redundancy	Relationships between positions	000 111 000 111 …

Lossless compression changes the representation while preserving the decoded sample values.

How much coding redundancy remains?

- Compare a code's average length with the symbol-entropy bound.

$$R_{\text{coding}} = L - H(X)$$

$$r_{\text{coding}} = 1 - \frac{H(X)}{L}$$

Absolute gap: bit/source symbol. Relative gap: dimensionless, often %. These compare codes under an independent-symbol model.

The same source, two code lengths

- The source entropy is unchanged when we choose a different code.

Code	L	L−H [bit/symbol]	Relative gap
Fixed-length	3	0.95976	31.992%
Probability-based	2.05	0.00976	0.476%

Here H=2.04024 bit/symbol. The relative gap above entropy is different from the saving relative to the original three-bit code.

What does the previous symbol tell us?

- Conditional entropy is the remaining uncertainty: H(current | previous) ≤ H(current).

Source	H(Xn)	H(Xn | Xn−1)
A: independent 000 / 111	1	1
B: alternating 000 111 …	1	0

Units: bit/source symbol. Conditioning cannot increase uncertainty on average. A repeating pattern may require more than one previous symbol.

Mutual information is the uncertainty removed

- Subtract the uncertainty remaining from the uncertainty before context.

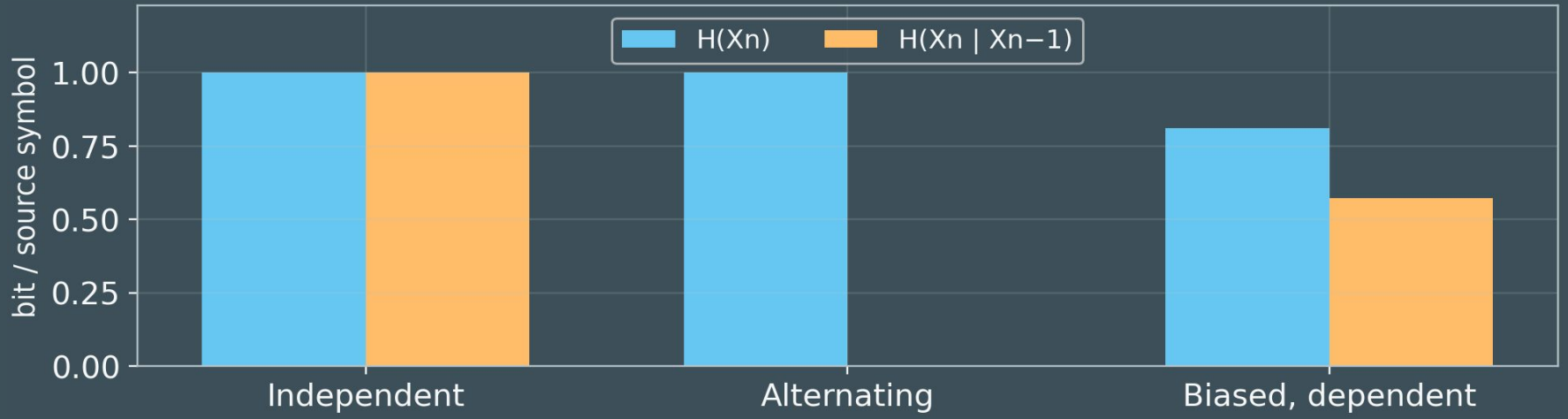
$$I(X_n; X_{n-1}) = H(X_n) - H(X_n \mid X_{n-1})$$

$$I_A = 1 - 1 = 0, \qquad I_B = 1 - 0 = 1$$

Report bits per adjacent pair. Mutual information is nonnegative. Zero adjacent mutual information does not rule out longer patterns.

Partial dependence gives intermediate values

- A biased source tends to repeat: P(111|000)=0.1 and P(111|111)=0.7.



Stationary P(000)=0.75: H≈0.811, Hconditional≈0.572, I≈0.239. The previous symbol helps, but does not determine the next.

Keep language-model concepts distinct

- The connection is predictability from context, not an equivalence between the operations.

Concept	Role
Tokenization	Defines the source symbols
Embeddings	Represent tokens as learned vectors
Attention	Combines representations using context
Conditional entropy	Quantifies remaining uncertainty

For “The cat sat on the …”, context changes next-token probabilities. Token IDs and attention weights are not information-content values.

Exercise: Redundancy and dependence

- Work through the notebook and complete its exercise.

  01.6_Redundancy.ipynb

- Distinguish coding overhead from sequential structure. Calculate conditional entropy, mutual information, and relative coding redundancy.
- Save the completed notebook and upload it to Moodle.

From probabilities to numerical relationships

- Interpret the three-bit source symbols as quantization levels: 000 → 0 and 111 → 7.

$$\rho_{X,Y} = \frac{\operatorname{cov}(X,Y)}{\sigma_X \sigma_Y}$$

$$-1 \leq \rho_{X,Y} \leq 1$$

The correlation coefficient is dimensionless and measures linear relationships. It is undefined if either standard deviation is zero.

Covariance compares paired deviations

- Positive contributions occur when both values lie on the same side of their respective means.

$$\bar{x} = \frac{1}{N}\sum_i x_i, \qquad \bar{y} = \frac{1}{N}\sum_i y_i$$

$$\operatorname{cov}(X, Y) = \frac{1}{N}\sum_i (x_i - \bar{x})(y_i - \bar{y})$$

Covariance depends on units and scale. Correlation removes that scale by dividing by the two standard deviations.

Worked calculation: N=3 paired values

- X=[1,2,3] and Y=[5,7,9] satisfy Y=2X+3.

$$\bar{x} = 2, \quad \bar{y} = 7, \quad \operatorname{cov}(X, Y) = 4/3$$

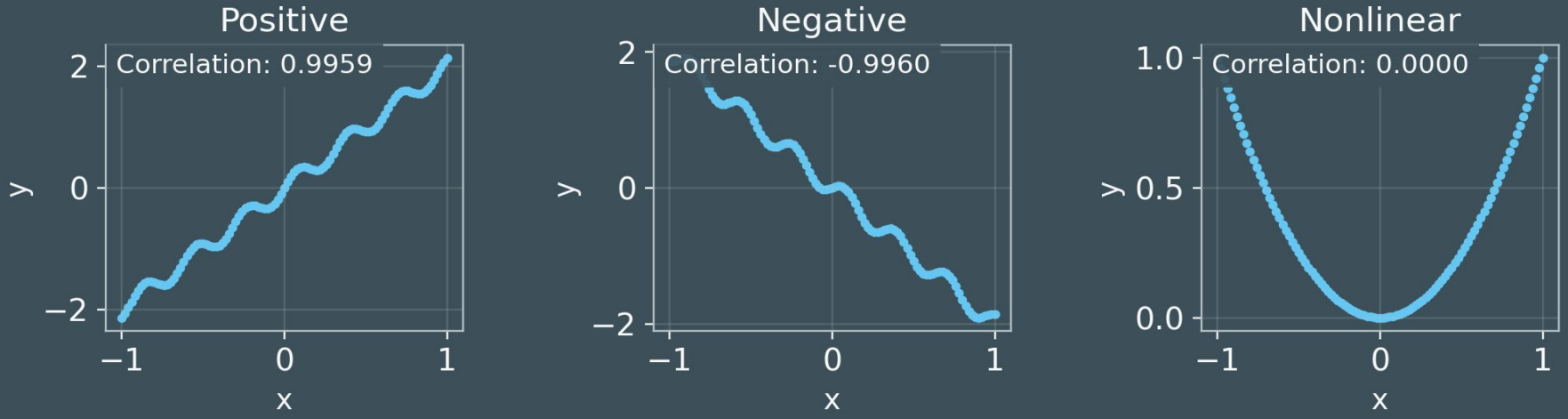
$$\sigma_X = \sqrt{2/3}, \quad \sigma_Y = \sqrt{8/3}$$

$$\rho = \frac{4/3}{\sqrt{2/3}\sqrt{8/3}} = 1$$

Each variance is the average squared deviation. Use the same N convention for covariance and both variances.

Zero correlation does not mean unpredictable

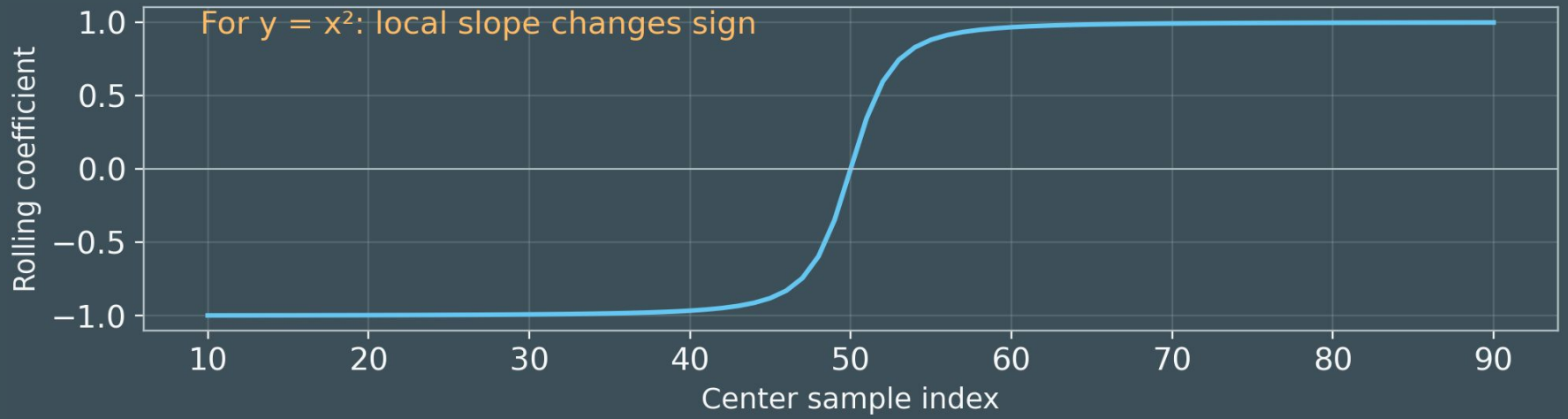
- A scatter plot shows paired values, while its label reports one overall coefficient.



For $y=x^2$ with symmetric x, positive and negative covariance contributions cancel. Yet x still determines y exactly.

Rolling correlation measures local relationships

- Recalculate the coefficient within each 21-pair window.



The window moves along sample position. This differs from changing the lag between signals.

Find a pattern at an unknown arrival time

- Cross-correlation compares a signal with shifted versions of a reference pattern.

$$R_{x,y}[k] = \sum_{n=0}^{N_x - 1} x[n]y[n-k]$$

$$k = -N_y + 1, \ldots, N_x - 1$$

Use zero outside valid indices. Positive k shifts y to the right. Raw sums depend on amplitude and overlap; they are not limited to [−1,1].

Cross-correlation: shift, multiply, sum

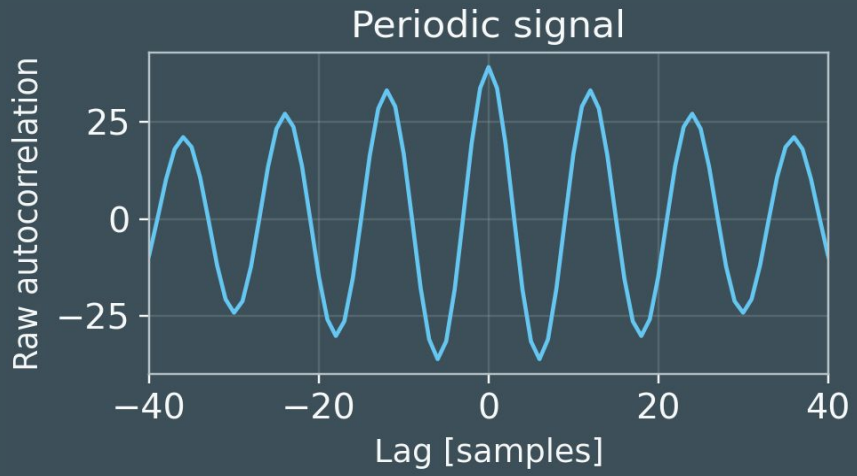
- For x=[1,2,3] and y=[1,2], evaluate every valid overlap.

Lag k	Overlapping products	Rxy[k]
−1	1·2	2
0	1·1 + 2·2	5
1	2·1 + 3·2	8
2	3·1	3

A receiver uses peaks to locate a preamble or estimate a delay. A negative result can indicate reversed polarity.

Autocorrelation looks for repetition

- Compare a signal with a shifted copy of itself.

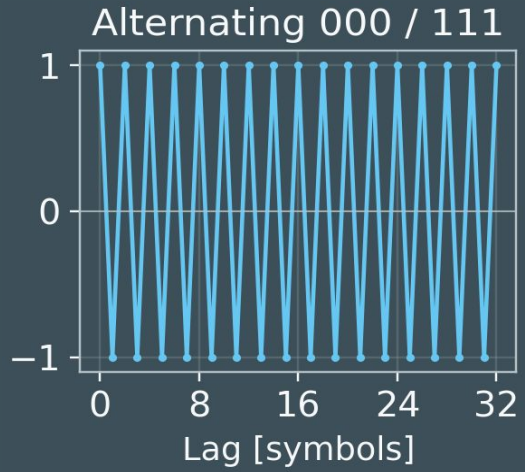


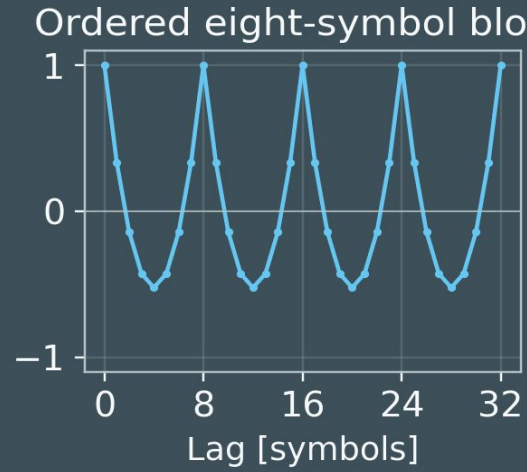


$R_{xx}[k]=\Sigma\, x[n]x[n-k]$. Repeated peaks near multiples of 12 reveal the period. The lag-zero peak alone also appears for noise.

Same symbol counts, different correlation

- Extend the short symbol examples to 4,096 symbols to estimate each lagged coefficient.







Repeat ordered patterns; extend shuffled examples by independent draws. Constant sequences are undefined: zero standard deviation.

Correlation and information: different measures

- The sign of correlation describes direction, not the amount of uncertainty.

Source	H	Hconditional	I	Correlation
A: independent	1	1	0	0
B: alternating	1	0	1	−1
C: biased, dependent	0.811	0.572	0.239	0.6

Alternation is completely predictable despite negative correlation. Mutual information is nonnegative; there is no general conversion from correlation to entropy.

Exercise: Correlation and lag

- Work through the notebook and complete its exercise.

  01.7_Correlation.ipynb

- Interpret the plots, calculate cross-correlation, identify a period from autocorrelation, and complete the short correlation function.
- Save the completed notebook and upload it to Moodle.

Exercise: Sampling theorem

- Work through the notebook and complete its exercise.

  01.8_Shannon_Nyquist_Sampling_Theorem.ipynb

- Explore aliasing and sinc reconstruction. Explain the Nyquist boundary, then choose a sensor sampling rate and calculate its data rate.
- Save the completed notebook and upload it to Moodle.

